Governing Equations and Representations of the Free Space Green's Tensor

T. M. Nestor

Research School of Earth Sciences, Australian National University

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2.1.2 The Governing Equations

Assuming that the depth below the free surface is measured by the $z-$coordinate, the forced elastic displacement $\boldsymbol{u} = (u_z, u_x, u_y)^T$ and traction vector $\boldsymbol{t} = (\tau_{zz}, \tau_{xz}, \tau_{yz})^T$ on a fixed depth plane (i.e. $z = const.$) with unit normal $\boldsymbol{e}_z$, are grouped into the 6-vector

$$\boldsymbol{b}(z,x,y) = \begin{bmatrix} u_z(z,x,y), & u_x(z,x,y), & u_y(z,x,y), \\ \tau_{zz}(z,x,y), & \tau_{xz}(z,x,y), & \tau_{yz}(z,x,y) \end{bmatrix}^T. \tag{1}$$

The governing equation for time harmonic elastic wave motion in an *isotropic medium* is then a set of 6 coupled partial differential equations (first order with respect to the depth coordinate) which may be written in the operator notation

$$\boldsymbol{\mathcal{L}}(\boldsymbol{x};\partial_z,\partial_x,\partial_y)\cdot\boldsymbol{b} \equiv \left[\partial_z - \boldsymbol{\mathcal{A}}(\boldsymbol{x};\partial_x,\partial_y)\right]\cdot\boldsymbol{b} = \boldsymbol{f} \tag{2}$$

where $\boldsymbol{b}$ is given in (1) and $\boldsymbol{\mathcal{A}}$ is defined as

$$\boldsymbol{\mathcal{A}}(\boldsymbol{x};\partial_x,\partial_y) = \begin{bmatrix} 0 & -\gamma\partial_x & -\gamma\partial_y & a & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & b & 0 \\ -\partial_y & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -\partial_x & -\partial_y \\ 0 & -\rho\omega^2 - \partial_x\zeta\partial_x - \partial_y\mu\partial_y & -\partial_x\chi\partial_y - \partial_y\mu\partial_x & -\partial_x\gamma & 0 & 0 \\ 0 & -\partial_y\chi\partial_x - \partial_x\mu\partial_y & -\rho\omega^2 - \partial_y\zeta\partial_y - \partial_x\mu\partial_x & -\partial_y\gamma & 0 & 0 \end{bmatrix} \tag{3}$$

and the material parameters in (3) are defined in Table 1.

This thesis is concerned with finding solutions of (2) when the material parameters in Table 1 are functions of (z,x) but are independent of the $y-$coordinate. This problem is sometimes called the $2\,1/2-$D problem because the wavefield may spread in $3-$D, but the heterogeneity is confined to $2-$D. We exploit the translational invariance of the medium along the $y-$coordinate by assuming that the displacement/traction field $\boldsymbol{b}(z,x,y)$ may be *synthesized* from its k_y plane

parameter	(ρ,α,β)-definition	(ρ,λ,μ)-definition
P	α	$\sqrt{\frac{\lambda+2\mu}{\rho}}$
S	β	$\sqrt{\frac{\mu}{\rho}}$
H	β	$\sqrt{\frac{\mu}{\rho}}$
γ	$\left(1-2\frac{\beta^2}{\alpha^2}\right)$	$\frac{\lambda}{\lambda+2\mu}$
ζ	$4\rho\beta^2\left(1-\frac{\beta^2}{\alpha^2}\right)$	$\frac{4\mu(\lambda+\mu)}{\lambda+2\mu}$
χ	$2\rho\beta^2\left(1-2\frac{\beta^2}{\alpha^2}\right)$	$\frac{2\lambda\mu}{\lambda+2\mu}$
a	$\frac{1}{\rho\alpha^2}$	$\frac{1}{\lambda+2\mu}$
b	$\frac{1}{\rho\beta^2}$	$\frac{1}{\mu}$

Table 1. The material parameter definitions

wave components as

$$\boldsymbol{b}(z,x,y) = \frac{1}{2\pi}\int_R \mathrm{d}k_y\, \boldsymbol{b}(z,x,k_y)\,\mathrm{e}^{ik_y y} \tag{4}$$

where $\boldsymbol{b}(z,x,k_y)$ is now governed by

$$\boldsymbol{\mathcal{L}}(z,x;\partial_z,\partial_x,ik_y)\cdot\boldsymbol{b}(z,x,k_y;\omega) \equiv \left[\partial_z - \boldsymbol{\mathcal{A}}(z,x;\partial_x,ik_y)\right]\cdot\boldsymbol{b}(z,x,k_y;\omega) = \tilde{\boldsymbol{f}} \tag{5}$$

with

$$\boldsymbol{\mathcal{A}}(z,x;\partial_x,ik_y) = \begin{bmatrix} 0 & -\gamma\partial_x & -i\gamma k_y & a & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & b & 0 \\ -ik_y & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -\partial_x & -ik_y \\ 0 & -\rho\omega^2-\partial_x\zeta\partial_x+\mu k_y^2 & -ik_y\partial_x\chi - ik_y\mu\partial_x & -\partial_x\gamma & 0 & 0 \\ 0 & -ik_y\chi\partial_x - ik_y\partial_x\mu & -\rho\omega^2+\zeta k_y^2-\partial_x\mu\partial_x & -ik_y\gamma & 0 & 0 \end{bmatrix} \tag{6}$$

and

$$\tilde{\boldsymbol{f}} = \int_R \mathrm{d}k_y\, \boldsymbol{f}(z,x,y)\,\mathrm{e}^{-ik_y y}. \tag{7}$$

In this work, we assume the the motion is driven by a point source in the $y=0$ plane, i.e. $\boldsymbol{f}(z,x,y) = \boldsymbol{F}(z,x)\delta(y)$, so that from (7) we have

$$\tilde{\boldsymbol{f}} = \int_R \mathrm{d}k_y\, \boldsymbol{F}(z,x)\delta(y)\,\mathrm{e}^{-ik_y y} = \boldsymbol{F}(z,x). \tag{8}$$

Due to the translational invariance of the medium in the $y-$coordinate, if the source plane is translated to $y=y_1$, then the spatial domain solution $\boldsymbol{b}(z,x,y;\omega)$ is translated to $\boldsymbol{b}(z,x,y-y_1;\omega)$. Consequently, it is straight forward to derive the solution appropriate to general source conditions, by translating the field for a point source in the $y=0$ plane, accordingly.

In (5) both ω and k_y are *passive parameters*. The space/time solution is constructed from a superposition of solutions with different values of ω and k_y as prescribed by the *inverse Fourier transforms* on frequency and on k_y. The main task remaining, is the construction the

transformed displacement/traction vector $\boldsymbol{b}(z, x, k_y; \omega)$ in (5). Often, we will not indicate the explicit dependence of $\boldsymbol{b}(z, x, k_y; \omega)$ on k_y and ω but write $\boldsymbol{b}(z, x)$ instead.

The representation (5) is most convenient when the dominant material variation is in the z coordinate direction, as we have singled out this coordinate for special treatment, allowing easy application of boundary and interface conditions on fixed depth planes. Also, the displacement $\boldsymbol{u}$ and depth plane traction $\boldsymbol{t}$ are accorded equal status. In the more traditional displacement formulation, the offset x and depth z coordinates appear within the equations in a symmetric form (due to isotropy) whereas the traction field must be derived from the displacement via the elastic analogue of *Hooke's law* which involves further spatial differentiations of $\boldsymbol{u}$.

We note in passing that (5) simplifies considerably in the special case where the motion is driven by a *line source*, along the $y-$axis so that we may set $k_y \equiv 0$ in (6) which then reduces to

$$\boldsymbol{\mathcal{A}}(z, x; \partial_x, 0) = \begin{bmatrix} 0 & -\gamma\partial_x & 0 & a & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & b & 0 \\ 0 & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -\partial_x & 0 \\ 0 & -\rho\omega^2 - \partial_x\zeta\partial_x & 0 & -\partial_x\gamma & 0 & 0 \\ 0 & 0 & -\rho\omega^2 - \partial_x\mu\partial_x & 0 & 0 & 0 \end{bmatrix}.$$

In this instance, (5) may be decomposed into its *in plane* equation

$$\frac{\partial}{\partial z}\begin{bmatrix} u_z \\ u_x \\ \tau_{zz} \\ \tau_{xz} \end{bmatrix} = \begin{bmatrix} 0 & -\gamma\partial_x & a & 0 \\ -\partial_x & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & -\partial_x \\ 0 & -\rho\omega^2 - \partial_x\zeta\partial_x & -\partial_x\gamma & 0 \end{bmatrix}\begin{bmatrix} u_z \\ u_x \\ \tau_{zz} \\ \tau_{xz} \end{bmatrix} \tag{9}$$

for the P/SV field and *out of plane* equation

$$\frac{\partial}{\partial z}\begin{bmatrix} u_y \\ \varnothing_{yz} \end{bmatrix} = \begin{bmatrix} 0 & b \\ -\rho\omega^2 - \partial_x\mu\partial_x & 0 \end{bmatrix}\begin{bmatrix} u_y \\ \varnothing_{yz} \end{bmatrix} \tag{10}$$

for the SH field. For the most part, we will concentrate on the solution of (5) because the source conditions needed for the applicability of (9) and (10) are too restrictive.

Consistent with the decomposition of the material properties, we *split* (6) into

$$\boldsymbol{\mathcal{A}}(\boldsymbol{x}; \partial_x, \partial_y; k_y) = \mathring{\boldsymbol{A}}(\boldsymbol{x}; \partial_x, \partial_y) + \Delta\boldsymbol{A}(\boldsymbol{x}; \partial_x, \partial_y) \tag{11}$$

where

$$\mathring{\boldsymbol{A}}(z,x;\partial_x,k_y) = \begin{bmatrix} 0 & -\mathring{\gamma}\partial_x & -i\mathring{\gamma}k_y & \mathring{a} & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & \mathring{b} & 0 \\ -ik_y & 0 & 0 & 0 & 0 & \mathring{b} \\ -\mathring{\rho}\omega^2 & 0 & 0 & 0 & -\partial_x & -ik_y \\ 0 & -\mathring{\rho}\omega^2 - \partial_x\mathring{\zeta}\partial_x + \mathring{\mu}k_y^2 & -ik_y\partial_x\mathring{\chi} - ik_y\mathring{\mu}\partial_x & -\partial_x\mathring{\gamma} & 0 & 0 \\ 0 & -ik_y\mathring{\chi}\partial_x - ik_y\partial_x\mathring{\mu} & -\mathring{\rho}\omega^2 + \mathring{\zeta}k_y^2 - \partial_x\mathring{\mu}\partial_x & -ik_y\mathring{\gamma} & 0 & 0 \end{bmatrix} \tag{12}$$

where $\mathring{\gamma}$ is the *reference medium* value of γ from Table 1 and similar definitions hold for the remaining *reference* material parameters in (12). Also,

$$\Delta\boldsymbol{A}(z,x;\partial_x,k_y) = \begin{bmatrix} 0 & -\Delta\gamma\partial_x & -i\Delta\gamma k_y & \Delta a & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta b & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta b \\ -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\Delta\rho\omega^2 - \partial_x\Delta\zeta\partial_x + \Delta\mu k_y^2 & -ik_y\partial_x\Delta\chi - ik_y\Delta\mu\partial_x & -\partial_x\Delta\gamma & 0 & 0 \\ 0 & -ik_y\Delta\chi\partial_x - ik_y\partial_x\Delta\mu & -\Delta\rho\omega^2 + \Delta\zeta k_y^2 - \partial_x\Delta\mu\partial_x & -ik_y\Delta\gamma & 0 & 0 \end{bmatrix} \tag{13}$$

with the material contrast parameters in (13) defined in Table 2.

contrast	definition
$\Delta\rho$	$\rho - \mathring{\rho}$
$\Delta\mu$	$\mu - \mathring{\mu}$
$\Delta\gamma$	$2\frac{\mathring{\beta}^2}{\mathring{\alpha}^2} - 2\frac{\beta^2}{\alpha^2}$
$\Delta\zeta$	$4\rho\beta^2\left[1 - \frac{\beta^2}{\alpha^2}\right] - 4\mathring{\rho}\mathring{\beta}^2\left[1 - \frac{\mathring{\beta}^2}{\mathring{\alpha}^2}\right]$
$\Delta\chi$	$2\rho\beta^2\left[1 - 2\frac{\beta^2}{\alpha^2}\right] - 2\mathring{\rho}\mathring{\beta}^2\left[1 - 2\frac{\mathring{\beta}^2}{\mathring{\alpha}^2}\right]$
Δa	$\frac{1}{\rho\alpha^2} - \frac{1}{\mathring{\rho}\mathring{\alpha}^2}$
Δb	$\frac{1}{\rho\beta^2} - \frac{1}{\mathring{\rho}\mathring{\beta}^2}$

Table 2. The material contrast parameters

Using the decomposition of (11) in the governing equation (5) gives

$$\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y)\cdot\boldsymbol{b}(z,x,k_y;\omega) = \tilde{\boldsymbol{f}} + \Delta\boldsymbol{A}(z,x;\partial_x,k_y)\cdot\boldsymbol{b}(z,x,k_y;\omega) \tag{14}$$

where the *reference medium* operator $\boldsymbol{L}$ is defined as

$$\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y) \equiv \left[\partial_z - \mathring{\boldsymbol{A}}\,(z,x;\partial_x,k_y)\right] \tag{15}$$

with $\mathring{\boldsymbol{A}}$ given in (12). Scrupulously indicating all the arguments of the various operators introduced in this thesis can be very cumbersome, so for example, we will often write $\boldsymbol{L}$ to mean $\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y)$ when no misunderstanding is likely.

Chapter 3: Representations of the Time Harmonic Elastic Wave Green's Tensor

We recall from Chapter 2 that the material properties of the propagation environment have been defined as the sum of a reference and residual component, i.e. $(\rho, \alpha, \beta) = (\mathring{\rho} + \Delta\rho, \mathring{\alpha} + \Delta\alpha, \mathring{\beta} + \Delta\beta)$. In this chapter, we will be concerned exclusively with uniform media so that residual properties $(\Delta\rho, \Delta\alpha, \Delta\beta)$ vanish and we have $(\rho, \alpha, \beta) = (\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$. To simplify notation, we discontinue the practice of writing $(\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$ to denote reference medium parameters and write (ρ, α, β) instead.

This chapter is concerned with the construction of the time harmonic elastic wave motion driven by a *lattice of multipolar point forces* $\boldsymbol{f}$ *superimposed on a homogeneous medium*. The homogeneous medium is characterized by the elastic parameters (ρ, α, β), which denote the density, compressional and shear speeds respectively, in an isotropic medium.

Each point force $\boldsymbol{f}$, has an assumed time dependence of the form $\exp(\epsilon t - i2\pi\nu t)$ where ν is the frequency and ϵ is a non-negative *growth* parameter. The *adiabatic growth term* $\mathrm{e}^{\epsilon t}$, evolves the wavefield from a state of quiescence in the infinite past, to its present value, following a path that ensures that the *wavefield is always in steady state equilibrium with the driving force* $\boldsymbol{f}$. This device has been employed by *Lighthill* (1960) to enforce the *radiation boundary conditions*. For convenience, we introduce the *complex angular frequency*

$$\omega \equiv 2\pi\nu + i\epsilon. \tag{16}$$

We can then synthesize *transient motion* from a discrete set of *forced motions*, sampled from the *Bromwich contour* in the complex frequency domain [e.g. *Chapman* (1978)] as

$$[\cdot]\,(t) = 2\,\mathrm{e}^{\epsilon t}\,\Re \int_0^{\nu_{max}} \mathrm{d}\nu\; \mathrm{e}^{-i2\pi\nu t}\,[\cdot]\,(2\pi\nu + i\epsilon) \tag{17}$$

where ν_{max} is the maximum frequency of the source field. Usually the growth parameter ϵ is chosen so that $5 \leq \epsilon T \leq 10$, where T is the length of the required time series (which is usually determined by trial and error). Using the complex angular frequency ω smoothes the frequency domain response, which enhances the conditioning of the problem, causing the complex frequency multiple scattering series to converge faster than the real frequency version. However, we must be aware that the process is equivalent to an analytic continuation, and is inherently unstable. The instability, is manifested by the sensitivity of (17) to errors in the frequency domain response. This is a particularly troubling issue, as the multiple scattering series is usually only summed to a relative error of 10^{-6} and it remains an open question whether we have secured sufficient accuracy to avoid the deleterious effects of the numerical instability inherent in (17). An excellent survey of numerical methods for the numerical evaluation of (17) is presented in *Davies & Martin* (1979).

3.1 The Free Space, Time Harmonic, Elastic Wave Green's Tensor

Given the 2-D vectors

$$\boldsymbol{x} \equiv (z, x), \quad \boldsymbol{x}' \equiv (z', x')$$

a particularly useful solution $\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y)$, satisfies (5) when the the material properties in Table 1 are homogeneous and the $y-$*Fourier transformed body force* $\boldsymbol{F}(\boldsymbol{x};\boldsymbol{x}')$ assumes the special form

$$\boldsymbol{F}(\boldsymbol{x};\boldsymbol{x}') = \boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}'),$$

with $\boldsymbol{I}_6$ the 6×6 *identity matrix, viz*

$$\boldsymbol{\mathcal{L}}\cdot\,\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y) = \boldsymbol{I}_6\delta(z-z')\delta(x-x'). \tag{18}$$

We can then use the linearity of (5) to pose the *particular solution* $\boldsymbol{b}_P$ of (5) as the *superposition integral*

$$\boldsymbol{b}_P = \int \mathrm{d}\boldsymbol{x}'\;\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y)\boldsymbol{f}(\boldsymbol{x}') \tag{19}$$

for an arbitrary body force $\boldsymbol{f}$. It will be convenient for the purpose of subsequent constructions, to interpret $\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y)$ of (19), as the *integral kernel* of the formal inverse

$$\left[\boldsymbol{\mathcal{L}}^{-1}\boldsymbol{w}\right](\boldsymbol{x}) \equiv \int \mathrm{d}\boldsymbol{x}'\;\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y)\cdot\boldsymbol{w}(\boldsymbol{x}') \tag{20}$$

for an arbitrary *test* vector-function $\boldsymbol{w}$, where $\boldsymbol{\mathcal{L}}$ is the governing operator in (5). Equation (20) is strictly formal, as the *null space* of $\boldsymbol{\mathcal{L}}$ is spanned by the set of *homogeneous* solutions and consequently, $\overset{\circ}{\boldsymbol{G}}$ is not unique. To get a *well posed* inverse, we invoke the *radiation condition* that *there are no homogeneous solutions of* (5) *due to sources at infinity*.

To solve for $\overset{\circ}{\boldsymbol{G}}$ in (18), or equivalently, to construct the inverse $\boldsymbol{\mathcal{L}}^{-1}$ in (20), we follow the classical *Fourier Transform* route. To begin, we transform the $x-$ dependence via $\mathcal{F}_x\,[\cdot](k_x) \equiv \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\;\exp\left(-i\,k_x x\right)\,[\cdot](x)$, where k_x is the horizontal component of the wavevector, so that the transverse *Fourier* transform of the governing equation (18) may be written as

$$\begin{aligned}\langle k_x|\boldsymbol{\mathcal{L}}|k_x{}'\rangle\langle k_x{}'|x\rangle\langle x|\;\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y)&\\ &\equiv \boldsymbol{L}(z,k_x,k_y)\;\overset{\circ}{\boldsymbol{G}}\,(z,k_x;z',x';k_y)\\ &\equiv \left[\partial_z - \boldsymbol{A}(k_x,k_y)\right]\;\overset{\circ}{\boldsymbol{G}}\,(z,k_x;z',x';k_y) = \boldsymbol{I}_6\delta(z-z')\,\frac{\exp(-ik_x x')}{\sqrt{2\pi}}\end{aligned} \tag{21}$$

where the *Fourier domain system matrix* $\langle k_x|\boldsymbol{\mathcal{A}}|k_x{}'\rangle = \delta(k_x-k_x{}')\boldsymbol{A}(k_x,k_y)$ is defined

$$\boldsymbol{A}(k_x,k_y) = \begin{bmatrix} 0 & -i\gamma k_x & -\gamma i\,k_y & a & 0 & 0\\ -ik_x & 0 & 0 & 0 & b & 0\\ -ik_y & 0 & 0 & 0 & 0 & b\\ -\rho\omega^2 & 0 & 0 & 0 & -ik_x & -ik_y\\ 0 & -\rho\omega^2+\zeta k_x^2+\mu k_y^2 & k_xk_y\,(\chi+\mu) & -ik_x\gamma & 0 & 0\\ 0 & k_xk_y\,(\chi+\mu) & -\rho\omega^2+\zeta k_y^2+\mu k_x^2 & -ik_y\gamma & 0 & 0 \end{bmatrix} \tag{22}$$

due to the *translational invariance of* $\boldsymbol{\mathcal{A}}$. The system operator $\boldsymbol{A}(k_x,k_y)$ of (22) is *quasi-Hamiltonian*

since

$$\boldsymbol{A}^T(-k_x,-k_y)\boldsymbol{J}_6 + \boldsymbol{J}_6\boldsymbol{A}(k_x,k_y) \equiv \boldsymbol{0}_6 \tag{23}$$

where $\boldsymbol{J}_6$ is the *canonically symplectic* array. The *diagonal array* $\boldsymbol{\Lambda}$ *of eigenvalues* of $\boldsymbol{A}(k_x,k_y)$ is defined as

$$\boldsymbol{\Lambda}(k_x,k_y) = i\,\mathrm{diag}\left[k_{z,P}(k_x,k_y), k_{z,S}(k_x,k_y), k_{z,H}(k_x,k_y), -k_{z,P}(k_x,k_y), -k_{z,S}(k_x,k_y), -k_{z,H}(k_x,k_y)\right] \tag{24}$$

where

$$k_{z,c}(k_x,k_y) = \begin{cases} \sqrt{\hat{K}_c^2(k_y) - k_x^2} & |k_x| < |\hat{K}_c(k_y)| \\ i\sqrt{k_x^2 - \hat{K}_c^2(k_y)} & |k_x| \geq |\hat{K}_c(k_y)| \end{cases}, \quad c \in \{P,S,H\}, \tag{25}$$

with

$$\hat{K}_c(k_y) = \begin{cases} \sqrt{(\omega/c)^2 - k_y^2} & |k_y| < |\omega/c| \\ i\sqrt{k_y^2 - (\omega/c)^2} & |k_y| \geq |\omega/c| \end{cases}, \quad c \in \{P,S,H\}. \tag{26}$$

We note that $k_{z,S} = k_{z,H}$ because $S = H = \beta$ from Table 1. We distinguish between $k_{z,S}$ and $k_{z,H}$, only to maintain a self-consistent notational convention. The system matrix $\boldsymbol{A}(k_x,k_y)$ of (22) has an equivalent *spectral representation*:

$$\boldsymbol{A}(k_x,k_y) \equiv \boldsymbol{D}_z(k_x,k_y)\boldsymbol{\Lambda}(k_x,k_y)\boldsymbol{D}_z^{-1}(k_x,k_y); \tag{27}$$

where the *eigen matrix* $\boldsymbol{D}_z(k_x,k_y)$ is defined as

$$\boldsymbol{D}_z(k_x,k_y) = \left[\ |+,P,k_x,k_y\rangle \quad |+,S,k_x,k_y\rangle \quad |+,H,k_x,k_y\rangle \quad |-,P,k_x,k_y\rangle \quad |-,S,k_x,k_y\rangle \quad |-,H,k_x,k_y\rangle\ \right] \tag{28}$$

with:

- the $|\pm,P,k_x,k_y\rangle$ eigen-vectors are the $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ $P-$waves

$$|\pm,P,k_x,k_y\rangle = \begin{bmatrix} \pm i\,k_{z,P}(k_x,k_y) \\ i\,k_x \\ i\,k_y \\ \rho\left(2\,\beta^2\,k_x^2 + 2\,\beta^2\,k_y^2 - \omega^2\right) \\ \mp 2\,\rho\,\beta^2\,k_x\,k_{z,P}(k_x,k_y) \\ \mp 2\,\rho\,\beta^2\,k_y\,k_{z,P}(k_x,k_y) \end{bmatrix} \epsilon_P \tag{29}$$

  with the normalization factor ϵ_P

- the $\left|\pm, S, k_x, k_y\right\rangle$ eigen-vectors are the $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ quasi-$SV-$waves

$$\left|\pm, S, k_x, k_y\right\rangle = \begin{bmatrix} i\,k_x \\ \mp i\,k_{z,S}(k_x,k_y) \\ 0 \\ \mp 2\,\rho\,\beta^2\,k_x\,k_{z,S}(k_x,k_y) \\ \rho\,\left(\omega^2 - 2\,\beta^2\,k_x^2 - \beta^2\,k_y^2\right) \\ -\rho\,\beta^2\,k_x\,k_y \end{bmatrix} \epsilon_S \tag{30}$$

  with the normalization factor ϵ_S

- the $\left|\pm, H, k_x, k_y\right\rangle$ eigen-vectors are the $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ quasi-$SH-$waves

$$\left|\pm, H, k_x, k_y\right\rangle = \begin{bmatrix} \mp k_y\,k_{z,H}(k_x,k_y) \\ -k_x\,k_y \\ \hat{K}_H^2 \\ 2\,i\,k_y\,\rho\,\left(\beta^2\,k_x^2 + \beta^2\,k_y^2 - \omega^2\right) \\ \mp 2\,i\,\rho\,\beta^2\,k_x\,k_y\,k_{z,H}(k_x,k_y) \\ \pm i\,k_{z,H}(k_x,k_y)\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right) \end{bmatrix} \epsilon_H \tag{31}$$

  with the normalization factor ϵ_H

We have called the eigen-vectors in (30) and (31) *quasi-SV* and *quasi-SH* waves respectively, because the corresponding vector fields are *solenoidal* and are *in the* (z,x) *plane* and *out of the* (z,x) *plane* respectively, when $k_y = 0$. However, our choice of the shear wavefield decomposition into quasi-inplane and quasi-out of plane components differs from the choice commonly used in seismology [e.g. *Prange & Toksöz* (1990)]. Unlike the conventional decomposition, the basis vectors (30) and (31) are *non-degenerate* when $k_x = 0$ and $k_y = 0$. The normalization factors ϵ_P, ϵ_S and ϵ_H in Eqns.(29)–(31) are chosen so that $\boldsymbol{D}_z^{-1}(k_x,k_y)$ takes a simple form.

Using the standard result that eigenspaces of $\boldsymbol{A}^T(k_x,k_y)$ annihilate eigenspaces of $\boldsymbol{A}(k_x,k_y)$ (and vice versa), we can avoid constructing $\boldsymbol{D}_z^{-1}(k_x,k_y)$ in (27) by explicit matrix inversion, since we use the *Hamiltonian symmetry* (23), to relate $\boldsymbol{D}_z^{-1}(k_x,k_y)$ to the eigen matrix of the *transposed* or *dual* system matrix $\boldsymbol{A}^T(k_x,k_y)$ defined as

$$\boldsymbol{A}^T(k_x,k_y) \equiv \boldsymbol{J}_6\boldsymbol{A}(-k_x,-k_y)\boldsymbol{J}_6$$

To begin, it follows from (24) and (25) that $\boldsymbol{\Lambda}(-k_x,-k_y) = \boldsymbol{\Lambda}(k_x,k_y)$, so that

$$\begin{aligned} \boldsymbol{D}_z(-k_x,-k_y)\boldsymbol{\Lambda}(k_x,k_y) &= \boldsymbol{A}(-k_x,-k_y)\boldsymbol{D}_z(-k_x,-k_y) \\ &= \boldsymbol{J}_6\boldsymbol{A}^T(k_x,k_y)\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y) \end{aligned}$$

which implies upon rearrangement that

$$\boldsymbol{A}^T(k_x,k_y)\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y) = -\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)\boldsymbol{\Lambda}(k_x,k_y).$$

Consequently, $\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)$ is an eigenbasis of the dual operator $\boldsymbol{A}^T(k_x,k_y)$, with $-\boldsymbol{\Lambda}(k_x,k_y)$ the corresponding eigenvalue array. From direct evaluation it follows that

$$\big(\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)\big)^T\boldsymbol{D}_z(k_x,k_y) = i\begin{bmatrix} \boldsymbol{0}_3 & \boldsymbol{V} \\ -\boldsymbol{V} & \boldsymbol{0}_3 \end{bmatrix} \tag{32}$$

where

$$\boldsymbol{V} = \text{diag}\left[\ 2\rho\omega^2 k_{z,P}(k_x)\epsilon_P^2,\quad 2\mu k_{z,S}(k_x)\hat{K}_S^2\epsilon_S^2,\quad 2\rho\omega^2 k_{z,H}(k_x)\hat{K}_H^2\epsilon_H^2\ \right]$$

and $\hat{K}_c, c \in \{P,S,H\}$ was given by (26), so that if we choose the normalization factors ϵ_P, ϵ_S and ϵ_H in Eqns.(29)–(31) according to

$$\epsilon_P = \frac{1}{\sqrt{2\rho\omega^2 k_{z,P}}},\quad \epsilon_S = \frac{\omega}{\beta\,\hat{K}_S\,\sqrt{2\rho\omega^2 k_{z,S}}},\quad \text{and}\quad \epsilon_H = \frac{1}{\hat{K}_H\,\sqrt{2\rho\omega^2 k_{z,H}}} \tag{33}$$

then (32) simplifies to

$$\big(\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)\big)^T\boldsymbol{D}_z(k_x,k_y) = i\boldsymbol{J}_6 \tag{34}$$

so that upon rearrangement of (34) we get

$$\boldsymbol{D}_z^{-1}(k_x,k_y) = -i\,\boldsymbol{J}_6\boldsymbol{D}_z^T(-k_x,-k_y)\boldsymbol{J}_6 \tag{35}$$

Equation (35) for the inverse is reminiscent of the *symplectic* structures found in *Hamiltonian dynamical systems* [e.g. *Goldstein* (1980)].

We now transform the $z-$dependence of (21) using the *Fourier transform*

$$\mathcal{F}_z\,[\cdot](k_z) \equiv \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dz\ \exp\left(-i\,k_z\,z\right)\ [\cdot](z),$$

to get

$$\tilde{\boldsymbol{L}}(\boldsymbol{k})\ \mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) \equiv \left[i\boldsymbol{I}_6k_z - \boldsymbol{A}(k_x,k_y)\right]\ \mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) = \boldsymbol{I}_6\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi}, \tag{36}$$

where $\boldsymbol{x}' = (z',x')$ and $\boldsymbol{k} = (k_z,k_x)$, so that

$$\mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) \equiv \left[i\boldsymbol{I}_6k_z - \boldsymbol{A}(k_x,k_y)\right]^{-1}\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi}. \tag{37}$$

Using the spectral representation (27) in (37), we get

$$\begin{aligned}\mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) &= \boldsymbol{D}_z(k_x,k_y)[i\,\boldsymbol{I}_6\,k_z\ -\ i\,\boldsymbol{\Lambda}]^{-1}\boldsymbol{D}_z^{-1}(k_x,k_y)\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi} \\ &= \boldsymbol{D}_z(k_x,k_y)[\boldsymbol{J}_6\,k_z\ -\ \boldsymbol{J}_6\,\boldsymbol{\Lambda}]^{-1}\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi} \\ &= \boldsymbol{D}_z(k_x,k_y)\begin{bmatrix} \boldsymbol{0}_3 & -\boldsymbol{W}^+ \\ \boldsymbol{W}^- & \boldsymbol{0}_3 \end{bmatrix}\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi}\end{aligned} \tag{38}$$

where

$$\boldsymbol{W}^{\pm} = \text{diag}\left[\ \tfrac{1}{(k_z\mp k_{z,P})},\quad \tfrac{1}{(k_z\mp k_{z,S})},\quad \tfrac{1}{(k_z\mp k_{z,S})}\ \right]$$

Applying the inverse Fourier transform $\boldsymbol{\mathcal{F}}^{-1}\left[\cdot\right](\boldsymbol{x}) \equiv \frac{1}{2\pi}\int_{R^2} \mathrm{d}\boldsymbol{k} \exp\left(i\boldsymbol{k}\cdot\boldsymbol{x}\right)\left[\cdot\right](\boldsymbol{k})$, it follows from (38), that $\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)$ may be written

$$\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) = \boldsymbol{\mathcal{F}}^{-1}\left[\mathring{\boldsymbol{G}}\left(\boldsymbol{k};\boldsymbol{x}';k_y\right)\right](\boldsymbol{x}) = \frac{1}{4\pi^2}\int_{R^2} \mathrm{d}\boldsymbol{k} \exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right] \times \boldsymbol{D}_z(k_x,k_y)\begin{bmatrix} \boldsymbol{0}_3 & -\boldsymbol{W}^+ \\ \boldsymbol{W}^- & \boldsymbol{0}_3 \end{bmatrix} \boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6 \tag{39}$$

3.1.1 *The Classical Up/Down, Plane Wave Green's Tensor*

We now evaluate the k_z integral in (39) using the calculus of residues [e.g. *Aki & Richards* (1980)], which is possible provided the z component of $(\boldsymbol{x}-\boldsymbol{x}')$ in (39) is non-zero, i.e. $(z-z') \neq 0$.

We close the k_z integration contour in the $\begin{smallmatrix}\text{upper}\\ \text{lower}\end{smallmatrix}$ half of the complex k_z-plane when $(z-z') \gtrless 0$, in so doing we pick up the

$$\Re(\partial\omega/\partial k_{z,c}) \gtrless 0$$

residue contributions, associated with $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ group velocity, to get

$$\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) = \frac{-i}{2\pi}\int_R \mathrm{d}k_x \exp\left[ik_x(x-x')\right] \times \boldsymbol{D}_z(k_x,k_y)\boldsymbol{Q}^z(z;z')\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6 \tag{40}$$

where $\boldsymbol{Q}^z(z;z')$ in (40), is defined as

$$\boldsymbol{Q}^z(z;z') = \begin{bmatrix} \boldsymbol{0}_3 & H(z-z')\boldsymbol{E}(k_x;z;z') \\ H(z'-z)\boldsymbol{E}(k_x;z;z') & \boldsymbol{0}_3 \end{bmatrix} \tag{41}$$

with

$$\boldsymbol{E}(k_x;z;z') = \begin{bmatrix} \mathrm{e}^{ik_{z,P}(k_x)|z-z'|} & 0 & 0 \\ 0 & \mathrm{e}^{ik_{z,S}(k_x)|z-z'|} & 0 \\ 0 & 0 & \mathrm{e}^{ik_{z,H}(k_x)|z-z'|} \end{bmatrix}, \tag{42}$$

and $H(\cdot)$ is the *Heaviside step function*. The imposition of the *radiation boundary conditions* employed here, follows *Lighthill* (1960). Alternatively, we may introduce *attenuation* to move the poles off the integration contour [e.g. *Aki & Richards* (1980)].

3.2 Discrete Plane Wave Representations of the Free Space Elastodynamic Green's Tensor

3.2.1 The z Splitting

Basic results concerning the discrete Fourier (DFT) approximation to the inverse Fourier transform over horizontal wavenumber of a scalar function, are summarized in BOX **3.2** and a simple extension of the basic theory to incorporate Filon quadrature, while exploiting the speed of the FFT algorithm is presented in BOX **3.3**. We will employ these results to develop a DFT approximation to (40). We assume the $N_k = 2M$ element *Fourier lattice*

$$\{k_j = j\Delta k_x \mid j = -M, \cdots, M-1\} \tag{43}$$

where $\Delta k_x = 2\pi j / L_x$ and L_x defines the *fundamental spatial period* of the representation. We may then approximate (40) by the *discrete plane wave sum*

$$\begin{aligned}\overset{\circ}{\boldsymbol{G}}{}^z(\boldsymbol{x};\boldsymbol{x}';k_y) = \frac{-i\,\Delta k_x}{2\pi} \sum_{j=-M}^{j=M-1} \\ \times \exp\left[ik_j(x-x')\right] \boldsymbol{D}_z(k_j,k_y)\boldsymbol{Q}^z(k_j;z;z')\boldsymbol{D}_z^T(-k_j,-k_y)\,\boldsymbol{J}_6\end{aligned} \tag{44}$$

The discrete representation (44) is equivalent to assuming that the transverse dependence of the wavefield is both *bandlimited* (since N_k is finite) and periodically repeated (since Δk_x is finite).

The next step will be to rewrite the discrete sum (44) as an equivalent matrix operation. To this end, we define the N_k-element diagonal, horizontal-wavevector array:

$$\mathbf{k}_x = \text{diag}\left[0, \Delta k_x, \cdots, (M-1)\Delta k_x, -M\Delta k_x, \cdots, -\Delta k_x,\right]; \tag{45}$$

the N_k-element diagonal, vertical-wavevector array

$$\mathbf{k}_{z,c} = \text{diag}\left[k_{z,c}(0), k_{z,c}(\Delta k_x), \cdots, k_{z,c}((M-1)\Delta k_x), k_{z,c}(-M\Delta k_x), \cdots, k_{z,c}(-\Delta k_x)\right], \quad c \in \{P,S,H\} \tag{46}$$

and the N_k-element *Fourier vector*

$$\boldsymbol{F}(x;\mathbf{k}_x) = \begin{bmatrix} 1 & ,\cdots, & 1 \end{bmatrix} \cdot \mathrm{e}^{i\,\mathbf{k}_x\,x}. \tag{47}$$

It is important to note that $\boldsymbol{F}(x;\mathbf{k}_x)$ as defined in (47) is an N_k- row vector, even though $\mathbf{k}_x$ is an N_k-diagonal matrix.

Using the newly defined arrays (45)–(47), we can rewrite the *discrete plane wave* approximation (44) to the 6×6 Green's tensor in the equivalent matrix form

$$\begin{aligned}\overset{\circ}{\boldsymbol{G}}{}^z(\boldsymbol{x};\boldsymbol{x}';k_y) = -i\,\left(\boldsymbol{I}_6 \otimes \boldsymbol{F}(x;\mathbf{k}_x)\right)\boldsymbol{D}_z(\mathbf{k}_x,k_y) \\ \times \boldsymbol{Q}^z(\mathbf{k}_x;z;z')\boldsymbol{D}_z^T(-\mathbf{k}_x,-k_y)\,\left(\boldsymbol{I}_6 \otimes \boldsymbol{F}^T(x';-\mathbf{k}_x)\right)\,\boldsymbol{J}_6\end{aligned} \tag{48}$$

where:
the *Kronecker product* $(\boldsymbol{I}_6 \otimes \boldsymbol{F}(x';\mathbf{k}_x))$ is a $6 \times 6N_k$ array, $\boldsymbol{F}^H(x;\mathbf{k}_x)$ is the *Hermitian transpose* of

the Fourier vector $\boldsymbol{F}(x;\mathbf{k}_x)$ defined in (47) (so that $\boldsymbol{F}^H(x;\mathbf{k}_x) \equiv \boldsymbol{F}^T(x;-\mathbf{k}_x)$);
the $(6N_k) \times (6N_k)$ matrix analogue of (28) is defined as

$$\boldsymbol{D}_z(\mathbf{k}_x,k_y) = \left[\ |+,P,\mathbf{k}_x,k_y\rangle \quad |+,S,\mathbf{k}_x,k_y\rangle \quad |+,H,\mathbf{k}_x,k_y\rangle \quad |-,P,\mathbf{k}_x,k_y\rangle \quad |-,S,\mathbf{k}_x,k_y\rangle \quad |-,H,\mathbf{k}_x,k_y\rangle\ \right] \tag{49}$$

in terms of

- the $6N_k \times N_k$ P-wave partition

$$|\pm,P,\mathbf{k}_x,k_y\rangle = \sqrt{\frac{\Delta k_x}{4\pi\rho\omega^2\mathbf{k}_{z,P}}} \begin{bmatrix} \pm i\,\mathbf{k}_{z,P} \\ i\,\mathbf{k}_x \\ i\,k_y\,\boldsymbol{I} \\ \rho\,\left(2\,\beta^2\,\mathbf{k}_x^2 + (2\,\beta^2\,k_y^2 - \omega^2)\boldsymbol{I}\right) \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_x\,\mathbf{k}_{z,P} \\ \mp 2\,\rho\,\beta^2\,k_y\,\mathbf{k}_{z,P} \end{bmatrix} \tag{50}$$

- the $6N_k \times N_k$ SV-wave partition

$$|\pm,S,\mathbf{k}_x,k_y\rangle = \frac{\omega}{\beta\hat{K}_S}\sqrt{\frac{\Delta k_x}{4\pi\rho\omega^2\mathbf{k}_{z,S}}} \begin{bmatrix} i\,\mathbf{k}_x \\ \mp i\,\mathbf{k}_{z,S} \\ \mathbf{0} \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_x\,\mathbf{k}_{z,S} \\ \rho\,\left((\omega^2 - \beta^2\,k_y^2)\boldsymbol{I} - 2\,\beta^2\,\mathbf{k}_x^2\right) \\ -\rho\,\beta^2\,k_y\,\mathbf{k}_x \end{bmatrix} \tag{51}$$

- the $6N_k \times N_k$ SH-wave partition

$$|\pm,H,\mathbf{k}_x,k_y\rangle = \frac{1}{\hat{K}_H}\sqrt{\frac{\Delta k_x}{4\pi\rho\omega^2\mathbf{k}_{z,H}}} \begin{bmatrix} \mp k_y\,\mathbf{k}_{z,H} \\ -k_y\,\mathbf{k}_x \\ \hat{K}_H^2\boldsymbol{I} \\ -2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_{z,H}^2 \\ \mp 2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_x\,\mathbf{k}_{z,H} \\ \pm i\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right)\,\mathbf{k}_{z,H} \end{bmatrix} \tag{52}$$

The $(6N_k) \times (6N_k)$ *plane wave propagator* $\boldsymbol{Q}^z(\mathbf{k}_x;z;z')$ in (48), is defined as

$$\boldsymbol{Q}^z(\mathbf{k}_x;z;z') = \begin{bmatrix} \mathbf{0}_{3N_k} & H(z-z')\boldsymbol{E}(\mathbf{k}_x;z;z') \\ H(z'-z)\boldsymbol{E}(\mathbf{k}_x;z;z') & \mathbf{0}_{3N_k} \end{bmatrix} \tag{53}$$

where the $(3N_k) \times (3N_k)$ *vertical phase shift array* is defined as

$$\boldsymbol{E}(\mathbf{k}_x;z;z') = \begin{bmatrix} \mathrm{e}^{i\mathbf{k}_{z,P}|z-z'|} & \mathbf{0}_{N_k} & \mathbf{0}_{N_k} \\ \mathbf{0}_{N_k} & \mathrm{e}^{i\mathbf{k}_{z,S}|z-z'|} & \mathbf{0}_{N_k} \\ \mathbf{0}_{N_k} & \mathbf{0}_{N_k} & \mathrm{e}^{i\mathbf{k}_{z,H}|z-z'|} \end{bmatrix},$$

in terms of the diagonal wavevector array $\mathbf{k}_{z,c}, c \in \{P, S, H\}$ of (46).

It will be convenient in later applications, to split (48) into the product

$$\mathring{\boldsymbol{G}}^z(\boldsymbol{x};\boldsymbol{x}';k_y) = \mathring{\boldsymbol{P}}^z(\boldsymbol{x};\boldsymbol{x}';k_y) \cdot \boldsymbol{J}_6$$

where $\boldsymbol{J}_6$ is the 6×6 *canonically symplectic* array and

$$\begin{aligned}\mathring{\boldsymbol{P}}^z(\boldsymbol{x};\boldsymbol{x}';k_y) = -i\ &(\boldsymbol{I}_6 \otimes \boldsymbol{F}(x;\mathbf{k}_x))\, \boldsymbol{D}_z(\mathbf{k}_x, k_y) \\ &\times \boldsymbol{Q}^z(\mathbf{k}_x; z; z')\boldsymbol{D}_z^T(-\mathbf{k}_x, -k_y)\ \left(\boldsymbol{I}_6 \otimes \boldsymbol{F}^T(x'; -\mathbf{k}_x)\right). \end{aligned} \tag{54}$$

might be called the *propagator*, for the want of a better name. The propagator is an entity in itself for most numerical applications considered here.

The representation (48) (or equivalently, (54)) is readily extensible to the plane stratified problem.

3.2.2 *The x Splitting*

Recalling the definition (28) for the columns of the eigen-matrix $\boldsymbol{D}_z$, (39) may be rewritten as

$$\begin{aligned}\mathring{\boldsymbol{G}}\ (\boldsymbol{x};\boldsymbol{x}';k_y) = &\sum_{c\in\{P,S,H\}} \frac{1}{4\pi^2}\int_{R^2} \mathrm{d}\boldsymbol{k} \exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right] \\ &\times \left[\frac{\left|-,c,k_x,k_y\right\rangle \otimes \left|+,c,-k_x,-k_y\right\rangle}{k_z + k_{z,c}(k_x)}\right. \\ &\qquad \left. - \frac{\left|+,c,k_x,k_y\right\rangle \otimes \left|-,c,-k_x,-k_y\right\rangle}{k_z - k_{z,c}(k_x)}\right] \boldsymbol{J}_6 \\ = &\mathring{\boldsymbol{G}}_P\ (\boldsymbol{x};\boldsymbol{x}';k_y) + \mathring{\boldsymbol{G}}_S\ (\boldsymbol{x};\boldsymbol{x}';k_y) + \mathring{\boldsymbol{G}}_H\ (\boldsymbol{x};\boldsymbol{x}';k_y) \end{aligned} \tag{55}$$

where $\otimes$ is the *tensor product*. We will ultimately evaluate the integrals in (55) using the residues at the simple poles, where

$$k_z^2 + k_x^2 + k_y^2 = \frac{\omega^2}{c^2}, \quad \text{for} \quad c \in \{P, S, H\}. \tag{56}$$

Hence, (55) may be replaced by an expression that is *equivalent at these poles*. Introducing the partial differential operators

$$\boldsymbol{L}_P = \begin{bmatrix} \partial_z \\ \partial_x \\ \partial_y \\ 2\mu\partial_{zz} - \frac{\lambda\omega^2}{\alpha^2} \\ 2\mu\partial_{xz} \\ 2\mu\partial_{yz} \end{bmatrix} \qquad \boldsymbol{L}_S = \frac{\omega}{\beta\hat{K}_S}\begin{bmatrix} \partial_x \\ -\partial_z \\ 0 \\ 2\mu\partial_{xz} \\ \mu(-\partial_{zz} + \partial_{xx}) \\ \mu\partial_{xy} \end{bmatrix} \qquad \boldsymbol{L}_H = \frac{1}{\hat{K}_H}\begin{bmatrix} \partial_{yz} \\ \partial_{xy} \\ \frac{\omega^2}{\beta^2} + \partial_{yy} \\ 2\mu\partial_{yzz} \\ 2\mu\partial_{xyz} \\ \mu(\frac{\omega^2}{\beta^2}\partial_z + 2\partial_{yyz}) \end{bmatrix} \tag{57}$$

$\mathring{\boldsymbol{G}}_c\ (\boldsymbol{x};\boldsymbol{x}';k_y)$ in (55) may be replaced with the *equivalent* expression

$$\begin{aligned}\mathring{\boldsymbol{G}}_c\ (\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{1}{8\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\frac{\boldsymbol{L}_c\otimes\boldsymbol{L}'_c\left(\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]\right)}{k_{z,c}(k_x)\left[k_z+k_{z,c}(k_x)\right]}\boldsymbol{J}_6 \\ &\quad - \frac{1}{8\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\frac{\boldsymbol{L}_c\otimes\boldsymbol{L}'_c\left(\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]\right)}{k_{z,c}(k_x)\left[k_z-k_{z,c}(k_x)\right]}\boldsymbol{J}_6, \\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad c\in\{P,S,H\}; \\ &= \frac{1}{4\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\boldsymbol{L}_c\otimes\boldsymbol{L}'_c\,\frac{\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]}{\hat{K}_c^2-k_z^2-k_x^2}\boldsymbol{J}_6 \end{aligned} \tag{58}$$

where we have used the algebraic relation (56) and the differential operator $\boldsymbol{L}'_c$ operates on the $\boldsymbol{x}'$ coordinate.

Assuming that $x \neq x'$ in (58) we may evaluate the k_x integral in that equation using the calculus of residues to get

$$\begin{aligned}\mathring{\boldsymbol{G}}_c\ (\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{-i}{2\pi\rho\omega^2}\int_R \mathrm{d}k_z\,\boldsymbol{L}_c\otimes\boldsymbol{L}'_c \\ &\quad\times\frac{\exp\left[ik_z(z-z')\right]\exp\left[ik_{x,c}(k_z,k_y)|x-x'|\right]}{2\,k_{x,c}(k_z,k_y)}\boldsymbol{J}_6,\end{aligned} \tag{59}$$

with

$$k_{x,c}(k_z,k_y) = \begin{cases}\sqrt{\hat{K}_c^2(k_y)\ -\ k_z^2} & |k_x|\ <\ |\hat{K}_c(k_y)| \\ i\sqrt{k_z^2\ -\ \hat{K}_c^2(k_y)} & |k_x|\ \geq\ |\hat{K}_c(k_y)|\end{cases} \tag{60}$$

and $\hat{K}_c(k_y)$ was given by (26). Using (59) in (55) we get

$$\begin{aligned}\mathring{\boldsymbol{G}}_c\ (\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{-i}{2\pi\rho\omega^2}\sum_{c\in\{P,S,H\}} \\ &\quad\times\int_R \mathrm{d}k_z\,\boldsymbol{L}_c\otimes\boldsymbol{L}'_c\,\frac{\exp\left[ik_z(z-z')\right]\exp\left[ik_{x,c}(k_z,k_y)|x-x'|\right]}{2\,k_{x,c}(k_z,k_y)}\boldsymbol{J}_6\end{aligned} \tag{61}$$

Using the definition of the partial differential operators $\boldsymbol{L}_c$ from (57) in (55) and performing the relevant spatial differentiations we get

$$\begin{aligned}\mathring{\boldsymbol{G}}\ (\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{-i}{2\pi}\int_R \mathrm{d}k_z\exp\left[ik_z(z-z')\right] \\ &\quad\times\boldsymbol{D}_x(k_z,k_y)\boldsymbol{Q}^x(k_z;x;x')\boldsymbol{D}_x^T(-k_z,-k_y)\,\boldsymbol{J}_6\end{aligned} \tag{62}$$

provided $x \neq x'$, where:

$$\boldsymbol{D}_x(k_z,k_y) = \left[\ \overline{|+,P,k_z,k_y\rangle}\ \ \overline{|+,S,k_z,k_y\rangle}\ \ \overline{|+,H,k_z,k_y\rangle}\ \ \overline{|-,P,k_z,k_y\rangle}\ \ \overline{|-,S,k_z,k_y\rangle}\ \ \overline{|-,H,k_z,k_y\rangle}\ \right] \tag{63}$$

where:

- the $\overline{|\pm, P, k_z, k_y\rangle}$ states are the $\begin{smallmatrix}\text{rightgoing}\\ \text{leftgoing}\end{smallmatrix}$ $P-$waves

$$\overline{|\pm, P, k_{x,P}, k_y\rangle} = \frac{1}{\sqrt{2\rho\omega^2 k_{x,P}}} \begin{bmatrix} i\,k_z \\ \pm i\,k_{x,P}(k_x, k_y) \\ i\,k_y \\ \rho\,\left(2\,\beta^2\,k_{x,P}^2(k_x, k_y) + 2\,\beta^2\,k_y^2 - \omega^2\right) \\ \mp 2\,\rho\,\beta^2\,k_{x,P}(k_x, k_y)\,k_z \\ -2\,\rho\,\beta^2\,k_y\,k_z \end{bmatrix} \tag{64}$$

- the $\overline{|\pm, S, k_z, k_y\rangle}$ states are the $\begin{smallmatrix}\text{rightgoing}\\ \text{leftgoing}\end{smallmatrix}$ quasi-$SV-$waves

$$\overline{|\pm, S, k_z, k_y\rangle} = \frac{\omega}{\beta\,\hat{K}_S\,\sqrt{2\rho\omega^2 k_{x,S}}} \begin{bmatrix} \pm i\,k_{x,S}(k_x, k_y) \\ -i\,k_z \\ 0 \\ \mp 2\,\rho\,\beta^2\,k_{x,S}(k_x, k_y)\,k_z \\ \rho\,\left(\omega^2 - 2\,\beta^2\,k_{x,S}^2(k_x, k_y) - \beta^2\,k_y^2\right) \\ \mp\rho\,\beta^2\,k_{x,S}(k_x, k_y)\,k_y \end{bmatrix} \tag{65}$$

- the $\overline{|\pm, H, k_z, k_y\rangle}$ states are the $\begin{smallmatrix}\text{rightgoing}\\ \text{leftgoing}\end{smallmatrix}$ quasi-$SH-$waves

$$\overline{|\pm, H, k_z, k_y\rangle} = \frac{1}{\hat{K}_H\,\sqrt{2\rho\omega^2 k_{x,H}}} \begin{bmatrix} -k_y\,k_z \\ \mp k_{x,H}(k_x, k_y)\,k_y \\ \hat{K}_H^2 \\ 2\,i\,k_y\,\rho\,\left(\beta^2\,k_{x,H}^2(k_x, k_y) + \beta^2\,k_y^2 - \omega^2\right) \\ \mp 2\,i\,\rho\,\beta^2\,k_{x,H}(k_x, k_y)\,k_y\,k_z \\ i\,k_z\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right) \end{bmatrix} \tag{66}$$

the *horizontal plane wave propagator* $\boldsymbol{Q}^x(k_z; x; x')$ in (62), is defined as

$$\boldsymbol{Q}^x(k_z; x; x') = \begin{bmatrix} \mathbf{0}_3 & H(x - x')\boldsymbol{E}(k_z; x; x') \\ H(x' - x)\boldsymbol{E}(k_z; x; x') & \mathbf{0}_3 \end{bmatrix} \tag{67}$$

with the *horizontal phase shift* array

$$\boldsymbol{E}(k_z; x; x') = \begin{bmatrix} \mathrm{e}^{ik_{x,P}(k_z)|x-x'|} & 0 & 0 \\ 0 & \mathrm{e}^{ik_{x,S}(k_z)|x-x'|} & 0 \\ 0 & 0 & \mathrm{e}^{ik_{x,H}(k_z)|x-x'|} \end{bmatrix},$$

and $H(\cdot)$ is the *Heaviside step function*.

For computational purposes, we must employ a *discrete plane wave* or *Fourier lattice* version of the wavenumber integral (62). The development closely parallels that of §3.2.1. We will now

derive a *discrete plane wave* version of (62), and we note that we will employ (62) only in the special case where $z = z'$ so that $\exp\left[i\bar{k}_j(z-z')\right] \equiv 1$ and consequently

$$\mathring{\boldsymbol{G}}\,(x;x';k_y) = \frac{-i}{2\pi}\int_R \mathrm{d}k_z\,\boldsymbol{D}_x(k_z,k_y) \times\ \boldsymbol{Q}^x(k_z;x;x')\boldsymbol{D}_x^T(-k_z,-k_y)\,\boldsymbol{J}_6, \quad z=z',\ x\neq x' \tag{68}$$

so there will be no *aliasing* or *wraparound* in the $z-$coordinate. This observation is crucial to the success in the intended application of $\mathring{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x}';k_y)$ from (68) to problems with a $z-$stratification in material properties.

Given the $\overline{N}_k = 2\overline{M}+1$ element *Fourier lattice*

$$\left\{\bar{k}_j = j\Delta k_z \mid j = -\overline{M},\cdots,\overline{M}\right\} \tag{69}$$

where $\Delta k_z = 2\pi j/L_z$ and L_z controls the *wavenumber sampling rate*, we may approximate (68) by the *discrete plane wave sum*

$$\mathring{\boldsymbol{G}}^x(x;x';k_y) = \frac{-i\,\Delta k_z}{2\pi}\sum_{j=-\overline{M}}^{j=\overline{M}}\boldsymbol{D}_x(\bar{k}_j,k_y) \times\ \boldsymbol{Q}^x(\bar{k}_j;x;x')\boldsymbol{D}_x^T(-\bar{k}_j,-k_y)\,\boldsymbol{J}_6 \tag{70}$$

The discrete representation (70) is equivalent to assuming that the $z-$dependence of the wavefield is *bandlimited* (since $\overline{M}$ is finite), which is certainly not the case in the limit $x \leftarrow x'$.

The next step will be to rewrite the discrete sum (70) as an equivalent matrix operation. To this end, we define the diagonal wave vector arrays:

$$\mathbf{k}_z = \mathrm{diag}\left[-\overline{M}\Delta k_z,\cdots,-\Delta k_z,0,\Delta k_z,\cdots,\overline{M}\Delta k_z\right]; \tag{71}$$

$$\mathbf{k}_{x,c} = \mathrm{diag}\left[k_{x,c}(-\overline{M}\Delta k_z),\cdots,k_{x,c}(-\Delta k_z),k_{x,c}(0),k_{z,c}(\Delta k_z),\cdots,k_{x,c}(\overline{M}\Delta k_z)\right] \tag{72}$$

and the $\overline{N}_k-$element *projection* vector

$$\overline{\boldsymbol{F}}(0) = \begin{bmatrix} 1 & \cdots & 1 & \cdots & 1 \end{bmatrix}. \tag{73}$$

We can rewrite the *discrete plane wave* approximation (70) to the 6×6 Green's tensor in the equivalent matrix form

$$\mathring{\boldsymbol{G}}^x(x;x';k_y) = -i\,\left(\boldsymbol{I}_6\otimes\overline{\boldsymbol{F}}(0)\right)\boldsymbol{D}_x(\mathbf{k}_z,k_y) \times\ \boldsymbol{Q}^x(\mathbf{k}_z;x;x')\boldsymbol{D}_x^T(-\mathbf{k}_z,-k_y)\,\left(\boldsymbol{I}_6\otimes\overline{\boldsymbol{F}}^T(0)\right)\,\boldsymbol{J}_6 \tag{74}$$

where:

the $(6\bar{N}_k) \times (6\bar{N}_k)$ matrix analogue of (63) is defined as

$$\boldsymbol{D}_x(\mathbf{k}_z, k_y) = \left[\overline{|+, P, \mathbf{k}_z, k_y\rangle} \quad \overline{|+, S, \mathbf{k}_z, k_y\rangle} \quad \overline{|+, H, \mathbf{k}_z, k_y\rangle} \quad \overline{|-, P, \mathbf{k}_z, k_y\rangle} \quad \overline{|-, S, \mathbf{k}_z, k_y\rangle} \quad \overline{|-, H, \mathbf{k}_z, k_y\rangle} \right] \tag{75}$$

in terms of

- the $6\overline{N}_k \times \overline{N}_k$ P-wave partition

$$\overline{|\pm, P, \mathbf{k}_z, k_y\rangle} = \sqrt{\frac{\Delta k_z}{4\pi\rho\omega^2 \mathbf{k}_{x,P}}} \begin{bmatrix} i\,\mathbf{k}_z \\ \pm i\,\mathbf{k}_{x,P} \\ i\,k_y\,\boldsymbol{I} \\ \rho\,\left(2\,\beta^2\,\mathbf{k}_{x,P}^2 + (2\,\beta^2\,k_y^2 - \omega^2)\boldsymbol{I}\right) \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_{x,P}\,\mathbf{k}_z \\ -2\,\rho\,\beta^2\,k_y\,\mathbf{k}_z \end{bmatrix} \tag{76}$$

- the $6\overline{N}_k \times \overline{N}_k$ SV-wave

$$\overline{|\pm, S, \mathbf{k}_z, k_y\rangle} = \frac{\omega}{\beta\hat{K}_S} \sqrt{\frac{\Delta k_z}{4\pi\rho\omega^2 \mathbf{k}_{x,S}}} \begin{bmatrix} \pm i\,\mathbf{k}_{x,S} \\ -i\,\mathbf{k}_z \\ \mathbf{0} \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_{x,S}\,\mathbf{k}_z \\ \rho\,\left((\omega^2 - \beta^2\,k_y^2)\boldsymbol{I} - 2\,\beta^2\,\mathbf{k}_{x,S}^2\right) \\ \mp\rho\,\beta^2\,k_y\,\mathbf{k}_{x,S} \end{bmatrix} \tag{77}$$

- the $6\overline{N}_k \times \overline{N}_k$ SH-wave partition

$$\overline{|\pm, H, \mathbf{k}_z, k_y\rangle} = \frac{1}{\hat{K}_H} \sqrt{\frac{\Delta k_z}{4\pi\rho\omega^2 \mathbf{k}_{x,H}}} \begin{bmatrix} -k_y\,\mathbf{k}_z \\ \mp k_y\,\mathbf{k}_{x,H} \\ \hat{K}_H^2\boldsymbol{I} \\ -2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_z^2 \\ \mp 2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_{x,H}\,\mathbf{k}_z \\ i\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right)\,\mathbf{k}_z \end{bmatrix} \tag{78}$$

where $\mathbf{k}_x$ and $\mathbf{k}_{z,c}$ were given in (72) and (71) respectively.

The $(6\bar{N}_k) \times (6\bar{N}_k)$ *plane wave propagator* $\boldsymbol{Q}^x(\mathbf{k}_z; x; x')$ in (74), is defined as

$$\boldsymbol{Q}^x(\mathbf{k}_z; x; x') = \begin{bmatrix} \mathbf{0}_{3\bar{N}_k} & H(x - x')\boldsymbol{E}(\mathbf{k}_z; x; x') \\ H(x' - x)\boldsymbol{E}(\mathbf{k}_z; x; x') & \mathbf{0}_{3\bar{N}_k} \end{bmatrix} \tag{79}$$

where the $(3\bar{N}_k) \times (3\bar{N}_k)$ *vertical phase shift array* is defined as

$$\boldsymbol{E}(\mathbf{k}_z; x; x') = \begin{bmatrix} \mathrm{e}^{i\mathbf{k}_{z,P}|x-x'|} & \mathbf{0}_{\overline{N}_k} & \mathbf{0}_{\overline{N}_k} \\ \mathbf{0}_{\overline{N}_k} & \mathrm{e}^{i\mathbf{k}_{z,S}|x-x'|} & \mathbf{0}_{\overline{N}_k} \\ \mathbf{0}_{\overline{N}_k} & \mathbf{0}_{\overline{N}_k} & \mathrm{e}^{i\mathbf{k}_{z,H}|x-x'|} \end{bmatrix},$$

in terms of the diagonal wavevector array $\mathbf{k}_{x,c}, c \in \{P, S, H\}$ of (72).

For computational efficiency, we split (74) into

$$\mathring{\boldsymbol{G}}^{x}(x; x'; k_y) = \mathring{\boldsymbol{P}}^{x}(x; x'; k_y)\boldsymbol{J}_6$$

where

$$\begin{aligned}\mathring{\boldsymbol{P}}^{x}(x; x'; k_y) = &-i\ \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}(0)\right) \boldsymbol{D}_x(\mathbf{k}_z, k_y) \\ &\times \boldsymbol{Q}^{x}(\mathbf{k}_z; x; x')\boldsymbol{D}_x^T(-\mathbf{k}_z, -k_y)\ \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right)\end{aligned} \tag{80}$$